Aarab Nishchal

+91-9546457837 | aarab.nishchal@gmail.com | aarab.me | linkedin.com/in/aarab-nishchal | github.com/aarabii

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS
Backend & Frameworks: Hono, Django, FastAPI, Node.js, Next.js, React, Astro, Tailwind CSS, Shadcn UI
Databases, Cloud & Edge: PostgreSQL, Prisma, Cloudflare Workers, Cloudflare Durable Objects, Cloudflare R2, AWS EC2, MongoDB, Supabase, Neon, ChromaDB, Convex
AI, LLMs & Automation: RAG, LangChain, Ollama, Gemini, OpenRouter, Groq, Whisper, VAPI, Trigger.dev, n8n, Transloadit, FFmpeg
Developer Tools: Git, GitHub, CI/CD, REST APIs, WebSockets, Tree-sitter, Zod, Zustand, Clerk, Postman, Jira

Experience

AI Engineer Intern July. 2026 – October 2026
Ascend HSI *Remote, Alaska, US*

- Architected the Setu internal attorney and client management system with a custom scheduling module to centralize legal case workflows.
- Integrated production LLM features and AI agents into internal SaaS pipelines to automate case management, CRM intelligence, and media placement.
- Built a dedicated content management and blogging platform for the internal content team to publish and manage articles directly.
- Refactored the Setu internal tool UI/UX to resolve workflow bottlenecks and improve system navigation for internal operations.

Software Developer Intern September 2024 – June 2026
Unstop *Remote, Delhi, IN*

- Engineered 200+ REST API endpoints and integrated n8n AI automation workflows for an in-house CMS to streamline content verification.
- Refactored and migrated the core database schema to clean legacy records and optimize query performance across the platform.
- Built an automated AI vendor assignment validation workflow to reduce manual creation and verification time by 60 to 70%.
- Resolved frontend UI/UX bugs and added white-label support with dynamic window-title configuration to improve vendor task flows.

Projects

Undercover | *Cloudflare Workers, Durable Objects, WebSockets, Astro, React* Live | GitHub

- Engineered a real-time multiplayer party game using WebSockets, Cloudflare Workers, and Astro with React to enable synchronized group gameplay.
- Offloaded WebSocket heartbeats to the Cloudflare Edge and cached alarms to reduce daily Durable Object request usage.
- Restricted storage writes to genuine game state mutations to ensure state persistence across 10-second Durable Object hibernations.

KodaArc | *TypeScript, Bun, Hono, PostgreSQL, Prisma 7, React 19, OpenTUI, Zod* Doc | GitHub

- Architected a four-package monorepo CLI coding assistant using React 19, OpenTUI, Hono, and PostgreSQL to automate local codebase edits and Git operations.
- Implemented conversation summarization, just-in-time context retrieval, and tool filtering to optimize LLM token consumption during long sessions.
- Built a Hono API server with Prisma 7 and shared Zod schemas to enforce end-to-end type safety across AI streaming and CRUD operations.

NextFlow | *Next.js 16, React Flow, Trigger.dev, Cloudflare R2, Transloadit, FFmpeg, Gemini, Neon* Live | GitHub

- Built a node-based visual AI workflow builder using Next.js 16 and React Flow, enabling users to construct and execute custom LLM and media pipelines.
- Engineered a Trigger.dev orchestration engine with tool calling to route graph nodes to Gemini for LLM inference and Transloadit/FFmpeg for media processing.
- Integrated Cloudflare R2 and Neon PostgreSQL via Prisma to persist processed audio/video assets and workflow execution states after every step.

TestIQ | *Python, LangChain, ChromaDB, Ollama, Tree-sitter* Doc | GitHub

- Built a local CLI RAG pipeline using Python, LangChain, and ChromaDB to analyze codebases, explain failing tests, and generate 60 passing unit tests.
- Engineered an AST-based validation loop using Tree-sitter and local Ollama inference to eliminate syntax errors before test output finalization.

Education

Kalinga Institute of Industrial Technology Bhubaneswar, India
Bachelor of Computer Science and Engineering *Sep. 2022 – Oct 2026*